

Compiler Design Assignment 3

Three-Address Code and Short-Circuit Code

Study **Section 6.6.1–6.6.5** of the textbook [**These sections are included in the Final tool**], Section 6.6.5 should be used specifically for the **second part** of every problem. The purpose is to modify the label assignments and control-flow structure so that **redundant goto instructions are avoided**.

Submission Format

- **Make a pdf of Handwritten submission and Submit the pdf through the GOOGLE FORM**
- Clearly separate the solution into **Part A** and **Part B** for your assigned problem.
- Show all intermediate three-address instructions clearly.

General Instructions

For all problems:

1. Assume C Language uses the following operators:
 - Arithmetic: $+$, $-$, $*$, $/$
 - Relational: $<$, $\leq$, $>$, $\geq$, $==$, $!=$
 - Boolean: $\&\&$, $||$, $!$
2. Assume $\&\&$ and $||$ may use **short-circuit evaluation**.
3. Arrays are one-dimensional unless otherwise stated.
4. **For the FIRST PART, generate the ordinary three-address control-flow code using explicit conditional and unconditional jumps.**
5. **For the SECOND PART, use the technique from Section 6.6.5 – Avoiding Redundant Gotos.**

Problem Assignment

Let L be the **last two digits of your student ID**

Calculate:

$$r = L \bmod 20$$

The assigned problem is determined as follows:

- If $r = 0$, you are assigned **Problem 20**.
- Otherwise, you are assigned **Problem r** .

Therefore, each student will solve **exactly one problem** from the 20 problems in Set Hard.

Example 1:

Student ID: **20231234**

$$L = 34$$

$$34 \bmod 20 = 14$$

Therefore:

Assigned Problem: Problem 14

Example 2:

Student ID: **20234560**

$$L = 60$$

$$60 \bmod 20 = 0$$

Therefore:

Assigned Problem: Problem 20

Important

- **Solve only the problem assigned to you.**
- **Your solution must contain both Part A and Part B.**
- In Part A, generate the ordinary three-address code.
- In Part B, transform the code using the **short-circuit/fall-through technique** of **Section 6.6.5**.
- **Students who submit a problem other than their assigned problem will not receive any marks for that problem.**

Problems

Problem 1

```
int A[20], B[20];
int i = 0;

while (i < 20) {
    if (A[i] > 0 && B[i] != 0 || A[i] < B[i])
        A[i] = A[i] + B[i];
    else
        B[i] = B[i] + 1;

    i = i + 1;
}
```

Problem 2

```
int A[30], B[30];
int i = 0;

while (i < 30) {
    if (!(A[i] < 0 || B[i] > 100 && B[i] > A[i]))
        B[i] = A[i] * 2;
    else
        A[i] = B[i] - 5;

    i = i + 1;
}
```

Problem 3

```
int A[25], B[25], C[25];
int i = 0;

while (i < 25) {
    if ((A[i] > B[i] && B[i] != 0) || C[i] == 10)
        C[i] = A[i] + B[i];
    else
        C[i] = C[i] - 1;

    i = i + 1;
}
```

Problem 4

```
int A[40], B[40];
int i = 0;

while (i < 40) {
    if (!(A[i] <= 10 || B[i] == 0) && A[i] <= B[i])
        A[i] = A[i] / 2;
    else
        B[i] = B[i] + A[i];

    i = i + 1;
}
```

Problem 5

```
int A[50], B[50], C[50];
int i = 0;

while (i < 50) {
    if (A[i] == 0 || !(B[i] > 5 && C[i] < 20))
        C[i] = C[i] + 2;
    else
        A[i] = A[i] - B[i];

    i = i + 1;
}
```

Problem 6

```
int A[20], B[20], C[30];
int i = 0;

while (i < 20) {
    if (A[i] >= 10 && !(B[i] < 5) || A[i] == C[i])
        A[i] = A[i] + 1;
    else
        B[i] = B[i] * 2;

    if (A[i] > B[i])
        i = i + 2;
}
```

Problem 7

```
int A[100], B[100], C[100];
int i = 0;

while (i < 100) {
    if ((A[i] != B[i] && C[i] > 0) || (A[i] == 0 && B[i] < C[i]))
        C[i] = C[i] + A[i];
    else
        C[i] = C[i] - B[i];

    i = i + 1;
}
```

Problem 8

```
int A[30], B[30];
int i = 0;

while (i < 30) {
    if ((A[i] > 0 || B[i] > 0) && A[i] != B[i])
        A[i] = A[i] + 5;
    else
        B[i] = B[i] + 5;

    i = i + 1;
}
```

Problem 9

```
int A[60], B[60], C[60];
int i = 0;

while (i < 60) {
    if (!(A[i] == B[i]) && (B[i] < C[i] || C[i] == 0))
        A[i] = C[i] - A[i];
    else
        B[i] = A[i] + C[i];

    i = i + 1;
}
```

Problem 10

```
int A[50], B[50];
int i = 0;

while (i < 50) {
    if (A[i] < B[i]) {
        if (B[i] < 100 && A[i] != 0)
            A[i] = A[i] + 10;
        else
            B[i] = B[i] - 10;
    }
    else {
        A[i] = A[i] - 1;
    }

    i = i + 1;
}
```

Problem 11

```
int A[40], B[40], C[40];
int i = 0;

while (i < 40) {
    if (A[i] > 0 && B[i] > 0 && C[i] > 0)
        C[i] = A[i] + B[i];
    else
        C[i] = A[i] - B[i];

    i = i + 1;
}
```

Problem 12

```
int A[40], B[40], C[40];
int i = 0;

while (i < 40) {
    if (A[i] == 0 || B[i] == 0 || C[i] == 0)
        A[i] = 0;
    else
        A[i] = A[i] + B[i] + C[i];

    i = i + 1;
}
```

Problem 13

```
int A[75], B[75];
int i = 0;

while (i < 75) {
    if ((A[i] >= 10 && A[i] <= 50) || !(B[i] >= 20))
        A[i] = A[i] + B[i];
    else
        B[i] = B[i] - A[i];

    i = i + 1;
}
```

Problem 14

```
int A[100], B[100], C[100];
int i = 0;

while (i < 100) {
    if (!(A[i] > B[i] && B[i] > C[i]) || A[i] != C[i])
        C[i] = C[i] + 1;
    else
        A[i] = A[i] - 1;

    i = i + 1;
}
```

Problem 15

```
int A[50], B[50], C[50];
int i = 0;

while (i < 50) {
    if ((A[i] != 0 && B[i] != 0) || !(C[i] > A[i]))
        A[i] = A[i] + C[i];
    else
        B[i] = B[i] + C[i];

    i = i + 1;
}
```

Problem 16

```
int A[30], B[30];
int i = 0;

while (i < 30) {
    if (A[i] > 0) {
        if (!(B[i] > 0 || A[i] > B[i]))
            A[i] = A[i] + B[i];
        else
            B[i] = B[i] + 1;
    }
    else {
        A[i] = 0;
    }

    i = i + 1;
}
```

Problem 17

```
int A[80], B[80], C[80];
int i = 0;

while (i < 80) {
    if ((A[i] < B[i] && B[i] < C[i]) || (A[i] == C[i]))
        C[i] = C[i] + 1;
    else
        C[i] = C[i] - 1;

    i = i + 1;
}
```

Problem 18

```
int A[100], B[100], C[100];
int i = 0;

while (i < 100) {
    if (!(A[i] == 0) && (B[i] != 0 || C[i] != 0))
        A[i] = A[i] + B[i] + C[i];
    else
        A[i] = 1;

    i = i + 1;
}
```

Problem 19

```
int A[60], B[60], C[60];
int i = 0;

while (i < 60) {
    if ((A[i] > 10 && B[i] <= 20) || !(C[i] != 0))
        A[i] = A[i] + 1;
    else
        B[i] = B[i] + 1;

    i = i + 1;
}
```

Problem 20

```
int A[100], B[100], C[100], D[100];
int i = 0;

while (i < 100) {
    if (!(A[i] > B[i] && B[i] > C[i]) || (C[i] <= 0)) {

        if (A[i] != D[i] || B[i] == 0)
            A[i] = A[i] + C[i];
        else
            B[i] = B[i] + D[i];

    }

    i = i + 1;
}
```